

NAYEEM HASAN

SENIOR SOFTWARE ENGINEER

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SUMMARY

Senior Backend Engineer with 8+ years building secure, high-performance distributed systems across EU-funded healthcare research, fintech, and e-commerce. Delivered backend infrastructure for FLUTE (active EU Horizon federated learning project); previously delivered full platform for TRUMPET (completed EU research project) - both spanning distributed hospital networks across Europe. Key wins: 2X traffic capacity, 65% faster API response, zero-downtime migration of 2M+ records. Core stack: Python (FastAPI, Django), PHP (Laravel), Azure, AWS. Available for full-time remote roles; async-friendly, CET/EST overlap comfortable.

PROFESSIONAL EXPERIENCE

Technovative Solutions LTD, Senior Software Engineer

May 2024 - May 2026

Tech stack: Python, FastAPI, Django, Celery, PostgreSQL, Docker, SPIRE, Envoy, Azure, GitHub, Pytest

- Engineered backend platform for two EU federated learning research projects (FLUTE & TRUMPET), building cloud platforms and on-premises Data Hub/Data Owner Nodes that enabled privacy-preserving AI workflows across distributed hospital partners without raw clinical data leaving local premises.
- Built role-based dashboards and RESTful APIs serving Researchers, Data Hub operators, and Governance/Admin roles across both platforms, with audit-ready access control, study approval workflows, and asynchronous task processing using Celery.
- Integrated HL7 FHIR and DICOM/Orthanc multimodal data pipelines standardising clinical datasets across hospital partners, supporting federated oncology workflows for NSCLC, HNC, and SBRT use cases.
- Implemented decentralised infrastructure security using mTLS with SPIRE/Envoy zero-trust architecture between cloud and on-premises hospital nodes, with Federated Learning Core integration enabling privacy-preserving model training across distributed sites on Azure.

Shajgoj Limited, Software Engineer

Mar 2022 - April 2024

Tech stack: PHP, Laravel, Redis, MySQL, Node.js, MongoDB, Apache Solr, Neo4j, Digital Ocean

- Redeveloped the core e-commerce backend from scratch, achieving 2X current traffic capacity with architecture capable of handling 4X peak load, and reducing average API response time by ~65%.
- Orchestrated migration of 2M+ orders, user records, and transactional data with zero downtime.
- Implemented a graph-based recommendation engine (Neo4j) and Apache Solr-powered search, improving product discovery and conversion.
- Delivered business-critical discount, cart, and order logic across desktop, mobile, and CRM platforms.
- Deployed and optimised cloud infrastructure on Digital Ocean, and implemented event-driven background job processing using Laravel Queues with Redis, reducing system load during peak traffic.

Walletmix Limited, Full Stack Developer

Feb 2021 - Feb 2022

Tech stack: PHP, Laravel, PostgreSQL, Redis, Vue.js, Docker, MySQL, GitHub

- Built a one-stop citizen engagement and service management platform for Dhaka North City Corporation (DNCC), serving 241K+ citizen records across active and in-process service workflows.
- Developed mobile APIs for Robi Alpha subscriber platform, ensuring full backward compatibility across existing client integrations.
- Maintained and extended a homegrown payment gateway platform providing payment infrastructure services to Bangladeshi businesses.

Japan Marketing and Consultancy Limited, Full Stack Developer

Jul 2019 - Jan 2021

Tech stack: PHP, Laravel, Vue.js, MySQL, Docker, AWS, Google Cloud, GitHub

- Collaborated across a 6-person cross-cultural development team (Bangladesh and Japan) coordinating across time zones to deliver a hotel booking platform targeting small-to-medium hotels in the Japanese market.
- Implemented server-side business logic and RESTful APIs supporting hotel registration, search, and booking workflows across desktop and mobile clients.
- Deployed and maintained the system on AWS at launch, optimising for reliability and operational efficiency.

Orange Business Development Limited, Junior Software Engineer

Oct 2018 - Jun 2019

Tech stack: PHP, Laravel, Vue.js, MySQL, GitHub

- Contributed to an a2i initiative platform streamlining website launches with essential onboarding information.
- Delivered features per project requirements across cross-functional teams.

Dream 71 Bangladesh Limited, Junior Software Engineer

Jan 2018 - Sep 2018

Tech stack: PHP, Laravel, Vue.js, MySQL

- Developed client-facing software as part of a collaborative backend team, following agile/scrum practices.

KEY PROJECTS

FLUTE: Federated Learning & Multi-Party Computation Platform (EU Research)

- Built the backend platform and API layer enabling privacy-preserving ML workflows across European hospitals.
- Integrated the Federated Learning Core into the platform infrastructure, enabling privacy-preserving model training across distributed hospital Data Hub nodes using PETs developed by research partners.
- Integrated HL7 FHIR pipelines to standardise multimodal clinical datasets (structured records and medical imaging via Orthanc/DICOM) across hospital partners, supporting federated oncology workflows for NSCLC, HNC, and SBRT use cases.
- Implemented and maintained decentralised infrastructure including mTLS security using SPIRE/Envoy architecture for zero-trust workload communication.

TRUMPET: TRUstworthy Multi-site Privacy Enhancing Technologies (EU Research)

- Built the cloud platform powering the Researcher, Data Owner, and Governance dashboards, with RESTful APIs, PostgreSQL backend, and role-based access control for multi-institutional federated workflows.
- Integrated the Federated Learning core aggregator into the cloud platform, enabling privacy-preserving model training across hospital Data Owner nodes without raw clinical data ever leaving local premises.
- Designed and implemented the Data Owner Node - an on-premises, privacy-by-design component covering dataset ingestion, FHIR-based data publishing, study approvals, and federated training execution against locally held clinical data.
- Validated the platform across three real-world clinical use cases (NSCLC, HNC, SBRT) involving hospital partners across Europe.

Shajgoj: E-Commerce Platform

- Led full backend and API development for mobile and CRM applications serving Bangladesh's leading beauty e-commerce brand.
- Built an extensive admin dashboard, recommendation engine, and multi-gateway payment integration.

TECHNICAL SKILLS

Languages: Python, PHP, JavaScript

Frameworks & Libraries: FastAPI, Django, Laravel, Node.js (Express.js), React.js, Vue.js

Databases: MySQL, PostgreSQL, MongoDB, Neo4j, Redis

Infrastructure & DevOps: Docker, GitHub Actions, CI/CD Pipelines

Architecture: Microservices, REST API, Event-Driven Architecture

Cloud Services: AWS, Google Cloud, Digital Ocean, Azure

Security: Zero-Trust Architecture (mTLS, SPIRE, Envoy), OAuth2

Healthcare & Interoperability: HL7 FHIR, DICOM

Testing Tools: Apache JMeter, Pytest

Version Control: Git

EDUCATION

B.S.C in Computer Science and Engineering (CSE)

Jan 2013 - Dec 2017

East West University

LANGUAGES

- **English:** Professional